

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>
The de Broglie wavelength (λ) of a particle is given by the equation $\lambda = h / \sqrt{2mE}$, where h is Planck's constant, m is the mass of the particle, and E is its kinetic energy. The mass of an electron is approximately 9.11×10^{-31} kg. The kinetic energy (E) of the electron is given as 1 eV (electron volt), which is equivalent to 1.602×10^{-19} J. Substituting these values into the equation, we get $\lambda = h / \sqrt{2mE} = (6.626 \times 10^{-34}) / \sqrt{2 \times 9.11 \times 10^{-31} \times 1.602 \times 10^{-19}} = 1.23 \times 10^{-7}$ cm
</think><answer> 1.23×10^{-7} cm</answer>

